

EA Credit: Enhanced Commissioning

This prerequisite applies to

- ▶ BD+C: New Construction (2-6 points)
- ▶ BD+C: Core & Shell (2-6 points)
- ▶ BD+C: Schools (2-6 points)
- ▶ BD+C: Retail (2-6 points)
- ▶ BD+C: Data Centers (2-6 points)
- ▶ BD+C: Warehouses & Distribution Centers (2-6 points)
- ▶ BD+C: Hospitality (2-6 points)
- ▶ BD+C: Healthcare (2-6 points)

INTENT

To further support the design, construction, and eventual operation of a project that meets the owner's project requirements for energy, water, indoor environmental quality, and durability.

REQUIREMENTS

NC, CS, SCHOOLS, RETAIL, DATA CENTERS, WAREHOUSES & DISTRIBUTION CENTERS, HOSPITALITY, HEALTHCARE

Implement, or have in place a contract to implement, the following commissioning process activities in addition to those required under EA Prerequisite Fundamental Commissioning and Verification.

Commissioning Authority Qualifications:

- ▶ The CxA must have documented commissioning process experience on at least two building projects with a similar scope of work. The experience must extend from early design phase through at least 10 months of occupancy;
- ▶ The CxA may be a qualified employee of the owner, an independent consultant, an employee of the design or construction firm who is not part of the project's design or construction team, or a disinterested subcontractor of the design or construction team.

Option 1. Enhanced Systems Commissioning (3-4 points)

Path 1: Enhanced Commissioning (3 points)

Complete the following commissioning (Cx) process activities for mechanical, electrical, plumbing, and renewable energy systems and assemblies in accordance with ASHRAE Guideline 0-2013 and ASHRAE Guideline 1.1-2007 for HVAC&R systems, as they relate to energy, water, indoor environmental quality, and durability.

The commissioning authority must do the following:

- ▶ Review contractor submittals.
- ▶ Verify inclusion of systems manual requirements in construction documents.
- ▶ Verify inclusion of operator and occupant training requirements in construction documents.
- ▶ Verify systems manual updates and delivery.
- ▶ Verify operator and occupant training delivery and effectiveness.
- ▶ Verify seasonal testing.
- ▶ Review building operations 10 months after substantial completion.
- ▶ Develop an on-going commissioning plan.

Include all enhanced commissioning tasks in the OPR and BOD.

OR

Path 2: Enhanced and Monitoring-Based Commissioning (4 points)

Achieve Path 1.

AND

Develop monitoring-based procedures and identify points to be measured and evaluated to assess performance of energy- and water-consuming systems.

Include the procedures and measurement points in the commissioning plan. Address the following:

- ▶ roles and responsibilities;
- ▶ measurement requirements (meters, points, metering systems, data access);
- ▶ the points to be tracked, with frequency and duration for trend monitoring;
- ▶ the limits of acceptable values for tracked points and metered values (where appropriate, predictive algorithms may be used to compare ideal values with actual values);
- ▶ the elements used to evaluate performance, including conflict between systems, out-of-sequence operation of systems components, and energy and water usage profiles;
- ▶ an action plan for identifying and correcting operational errors and deficiencies;
- ▶ training to prevent errors;
- ▶ planning for repairs needed to maintain performance; and
- ▶ the frequency of analyses in the first year of occupancy (at least quarterly).

Update the systems manual with any modifications or new settings, and give the reason for any modifications from the original design.

AND/OR

Option 2. Building Enclosure Commissioning (2 points)

Fulfill the requirements in EA Prerequisite Fundamental Commissioning and Verification as they apply to the building's enclosure in addition to mechanical and electrical systems and assemblies.

Complete the following commissioning (Cx) process activities for the building's thermal envelope in accordance with ASHRAE Guideline 0-2013 and ASTM E2947-16: Standard Guide for Building Enclosure Commissioning, as they relate to energy, air and water tightness, indoor environmental quality, and durability.

The qualified independent member of the design or construction team responsible for building enclosure commissioning must complete the following:

- ▶ Review contractor submittals.
- ▶ Verify inclusion of systems manual requirements in construction documents for enclosure systems.
- ▶ For specialty enclosure systems with controls and automation:
 - Verify inclusion of operator and occupant training requirements in construction documents.
 - Verify systems manual updates and delivery.
 - Verify operator and occupant training delivery and effectiveness.
 - Verify seasonal testing.
 - Review building operations 10 months after substantial completion.
- ▶ Develop an on-going enclosure commissioning plan for maintenance, renewal, and revitalization cycles.

Data Centers only

Projects that select Option 1 must complete the following commissioning process.

For small projects with peak cooling loads less than 2,000,000 Btu/h (600 kW), or a total computer room peak cooling load less than 600,000 Btu/h (175 kW), the CxA must perform the following activities:

- ▶ conduct at least one commissioning verification review of the owner's project requirements, basis of design, and design documents before mid-construction documents development;
- ▶ back-check the review comments in all subsequent design submissions; and
- ▶ conduct an additional full verification review at 95% completion of the design documents and basis of design.

For projects with peak cooling loads 2,000,000 Btu/h (600 kW) or more, or a total computer room peak cooling load 600,000 Btu/h (175 kW) or more, the CxA must conduct at least three verification reviews of the basis of design:

- ▶ one verification review of design documents before the start of design development;
- ▶ one verification review of design documents before midconstruction documents; and
- ▶ one final verification review of 100% complete design documents, verifying achievement of the owner's project requirements and adjudication of previous review comments.

GUIDANCE

Refer to the LEED v4 reference guide, with the following additions and modifications.

Behind the Intent

Beta Update

Updated referenced standards. Minor text revisions to Option 2. Building Enclosure Commissioning intend to clarify implementation for project teams.

Commissioning Authority criteria are expanded to include an employee of the design or construction firm who is not part of the project's design or construction team, or a disinterested subcontractor of the design or construction team.

Further Explanation

Building Enclosure Commissioning Basics

Refer to the LEED v4 reference guide, with the following modification:

- ▶ See ASTM E2947 - 16: Standard Guide for Building Enclosure Commissioning and ASTM E2813-12: Standard Practice For Building Enclosure Commissioning for additional guidance regarding envelope systems testing equipment and procedures.

Referenced Standards:

- ▶ ASHRAE Guideline 0-2013, The Commissioning Process
- ▶ ASHRAE Guideline 1.1-2007, HVAC&R Technical Requirements for the Commissioning Process
- ▶ ASTM E2947 - 16: Standard Guide for Building Enclosure Commissioning
- ▶ ASTM E2813-12: Standard Practice for Building Enclosure Commissioning

Connection to Ongoing Building Performance

- ▶ LEED O+M EA credit Energy Performance: Enhanced building commissioning expands the fundamental commissioning process to provide further oversight and verification of mechanical systems, which ensures ongoing building quality control and operations and may help improve the project's energy performance score. Monitoring-based commissioning is a powerful strategy for tracking building performance in real time. Building enclosure commissioning provides quality assurance for enclosure systems design and installation and provides the foundation for energy efficiency throughout the building life cycle.